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Installation

 WARNING

Prepare the installation site and clearly designate it! All unauthorised personnel must be restricted from the installation area. Only trained logistics specialists are to be permitted in the installation area.


 WARNING

When the nature of a task is such that it cannot be performed at floor level and a ladder or other suitable means is required, such means must not be used in contravention to prescribed safety practices.



After the palletised machine is placed near the installation site (reference the layout drawing for correct placement), unpack the machine as follows:

1. Remove any wooden crating (if applicable).
2. Remove any shrink wrap or packing films. Be careful not to damage the machine surface finishes.
3. Inspect the machine and incidental items for damage that may have occurred during transport. Report all damage to the manufacturer or to the manufacturer's agent immediately.
4. Create a list of the parts present in the packing and compare it with the packing list. Report all missing items to **BluePrint Automation** or **BluePrint Automation's** agent immediately.
5. Remove all of the shipping braces that constrain the motion of swinging parts and/or that secure the machine to the shipping pallet.
6. Remove all incidental items shipped with the machine.
7. Remove the machine from the pallet with a forklift.

User Manual

 WARNING

Never lift unbalanced loads. Never lift loads heavier than that for which the device is rated. Never lift loads while persons are moving in the area below. Never use the device for tasks such as pulling, dragging fastened loads, extracting guided elements, etcætera. Never position the balance plate vertically or on the side during the transport, movement, storage, and measurement phases. It is forbidden for personnel to ride on the transported load or to attempt mounting the transport equipment while it is in use or once the load has been applied. Never use the lifting device to lift and transport persons or objects other than the loads for which the device was designed. Use of undersized lifting equipment could result in damage to the machine or personal injury.



Be sure that the lift levers are long enough to extend past the barycentric axis of the machine.

8. Position the machine at the intended installation site.
9. With the forklift bearing the load of the machine, adjust the levelling feet so that the machine is level.
10. Lower the fork and verify that the machine is level; adjust as necessary.
11. Make sure the floor is suitable to support the weight of the system.
12. Dispose of all packaging material following applicable regulations.

Packing materials may vary:

- Stretch wrap: Linear low-density polythene.
- Underlay: Polythene sheet.

Space and Clearance

The minimum floor space and clearances required to operate comfortably on and around the machine are indicated on the layout drawing. Make sure space is allowed for opening panels and accessibility.

NOTICE

*This equipment supplied by **BluePrint Automation** is designed for a fixed location unless otherwise specified and should be securely anchored to prevent walking or moving in compliance with OSHA 1910.212(b).*



WARNING

- *The floor must not be subject to deterioration over time. The floor must be non-slip and even.*
- *If the machine has feet, never over-extend the feet. This can cause the machine to become unstable and topple.*
- *Make sure the machine guards are sufficiently close to the floor, even if the floor is uneven. These guards must always prevent that persons reach into the machine.*
- *If you connect the in-feed or out-feed conveyor of the machine to another machine, make sure that guarding prevents a person from reaching hazardous locations.*
- *Be careful when you adjust the height of the in-feed belt. Its support has sharp edges.*

The area around the machine must have enough space for the operator to go around the machine and do the normal operating procedures.

It must be possible to fully open the protective covers and the doors of the control cabinet. For this reason, the minimum distance to a wall or a potential obstacle must be approximately 2 metres.

To make cleaning of the machine easier and to prevent hindrance, make sure no loose cables are on the floor.

The capacity of the installed machine is given on the identification plates. For the main power supply, adhere to the standard instructions for cable thickness and sheathing.

For the correct operation of the pneumatic components in the machines, the capacity of the compressor must be the same as the air consumption of the machines of **BluePrint Automation**.

Electrical Installation

The main electrical cabinet is designed to work in standard indoor industrial operating environments, not exposed to the weather.

The following conditions are required for correct installation and operation:

- **+5...+32 °C with cooling fans installed in the main electrical cabinet**
- **+5...+45 °C with optional air conditioners in the main electrical cabinet**
- **20...70% relative humidity, without condensation effect.**

Electrical Service Connexion

Reference the electrical schematics. In addition, any machine sub-assemblies shipped loose will require restoration of their service connexions. The following tasks must be performed under the direction of a qualified electrical technician. In making these connexions, observe the following:

- Delegate appropriate personnel. Ensure these personnel are qualified for the task and are coordinated by an electrical engineer.
- Arrange the necessary ladders and ensure that they comply with the relevant safety laws in force. The ladders must be free-standing, have an adequate work platform to permit the technician to work freely with both hands, and must not under any circumstance rest against the equipment.
- Verify that all machine movement areas are clear of cables or electrical wires.
- Synchronise the work of the whole team for every operation.
- Take appropriate precautionary measures in those tasks that may be dangerous.
- The ends of the cables are marked with different codes. Use the components list and layout diagrams to match these codes on the machine and in the cabinet.
- While connecting and disconnecting the receptacles and plugs, keep the plugs aligned radially and axially to the receptacles to avoid damaging the contacts.



Only qualified persons shall be permitted to make the main electrical connexion.



The end-user's main power switch for the power lines to be installed must be switched off and locked out in accordance with the regulations in force in the user's country before making the electrical connexions in the main cabinet.



Electrical Supply: Because our clients have different requirements, gland access holes for installation of the main electrical power are not provided. Be certain wire gauge, cable gland fittings, and insulation meet all applicable code requirements. The power source must have a breaker in a lockable enclosure.

This connexion is made at the terminals of the main switch in the cabinet. Use proper tools when passing the cable through the cable gland. A cable of cross-section suitable to the installed power (see the electrical drawings) must be employed. Make sure that the values of the machine power, frequency, and voltage (see the plate on the main cabinet) correspond to the values of the electrical grid. The cable must enter through the cable gland fitting and must be connected to the main switch situated on the inside of the cabinet. The connexion is to be effected at the three terminals **L1-L2-L3** for the three-phase line and the **PE** terminal for the external ground conductor. In case the neutral conductor is present, it should be connected to the special terminal marked **N**.

Once these connexions are made, the cabinet becomes operative and the normal safety regulations are in effect:

- Admittance to the machinery is restricted to authorised personnel only.
- The doors must remain closed, except for maintenance or repairs with the main switch off and locked.

Prolonged Storage

If the machine, still packed, is expected not to be placed into operation for a prolonged period, we recommend that it be stored in a dry place (<55% humidity), with temperatures falling between +5 °C and +35 °C.

If the machine has already been unpacked and is still divided in sections, we suggest that it is stored in a dry place, on a proper base (if possible, the skid of the original packing).

If the machine has been assembled, it should be protected with suitable coverings. Moreover, protect any delicate apparatus (e.g., monitors, keyboards, displays etc.). Make sure that any outer exposed corrodible mechanical parts are protected. Cover the machine with a sheet that lets the inner, residual humidity evaporate.

Periodically check the status of the machine and of the visible parts to avoid the formation of superficial oxidation.

OEM Warranty:

- If it is decided to store the equipment for a prolonged period of time, before the equipment is put in operation, please contact the OEM with the serial numbers to ask for an extended warranty.

Oil In gear-heads:

- Contact the gear-head manufacturer for recommendation on oil drainage, if applicable.

PLC Battery and Memory Card:

- Follow recommendation as per Rockwell AB Micro820 controllers – Non-volatile Memory Card.



Machine Disposal

Once the equipment reaches the end of its useful life, it must be disposed of properly.



All hazards remain present until the packaging line is completely dismantled, disposed of or recycled.

Before disposal, the machine must be disassembled and the individual parts must be split into the following groups:

- Aluminium.
- Steel.
- Plastics.
- Electric motors.
- Control equipment and switchgear.
- Electric cables.

Sort all materials classified as chemical waste, such as batteries, oil, lubricants and certain electric components. Dispose of these materials as chemical waste.

All materials must be disposed of in accordance with the local environmental rules and regulations.

Possible residual risks during dismantling and removal:

- Disconnecting the power supply.
- Disassembly of machine parts.
- Lifting machine (parts).
- Transport of the machine (parts).
- Loading / unloading.
- Toxic or harmful chemicals.